

**** Basic C Programming Questions ****

1. Write a C program to display your name on the screen.

Ans: -

```
#include <stdio.h>
#include <conio.h>
void main()
{
    clrscr();

    printf("My Name is Subham Kumar Ray");

    getch();
}
```

Output: - My Name is Subham Kumar Ray

2. Write a C program to display "Welcome to C Programming".

Ans: -

```
#include <stdio.h>
#include <conio.h>
void main()
{
    clrscr();
    printf("Welcome to C Programming");
    getch();
}
```

Output: - Welcome to C Programming

3. Write a C program to display your name, age and city.

Ans: -

```
#include <stdio.h>
#include <conio.h>
void main()
{
    clrscr();
    printf("Name: Subham Kumar Ray\n");
    printf("Age: 21\n");
    printf("City: Hazaribagh");
    getch();
}
```

Output: - Name: Subham Kumar Ray
Age: 21
City: Hazaribagh

4. Write a C program to input two integers and find their sum.

Ans: -

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int a, b, sum;
    clrscr();
    printf("Enter two integers: ");
    scanf("%d%d", &a, &b);
    sum = a + b;
    printf("Sum = %d", sum);
    getch();
}
```

Output: - Enter two integers: 10 20

Sum = 30

5. Write a C program to input two numbers and find their difference.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int a, b, difference;
clrscr();
printf("Enter two numbers: ");
scanf("%d%d", &a, &b);
difference = a - b;
printf("Difference = %d", difference);
getch();
}

Output: - Enter two numbers: 50 20
Difference = 30

6. Write a C program to input two numbers and find their product.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int a, b, product;
clrscr();
printf("Enter two numbers: ");
scanf("%d%d", &a, &b);
product = a * b;
printf("Product = %d", product);
getch();
}

Output: - Enter two numbers: 10 5
Product = 50

7. Write a C program to input two numbers and find their division result.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
float a, b, result;
clrscr();
printf("Enter two numbers: ");
scanf("%f%f", &a, &b);
result = a / b;
printf("Division = %.2f", result);
getch();
}

Output: - Enter two numbers: 10 4
Division = 2.50

8. Write a C program to input two integers and find the remainder.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int a, b, remainder;

```

clrscr();
printf("Enter two integers: ");
scanf("%d%d", &a, &b);
remainder = a % b;
printf("Remainder = %d", remainder);
getch();
}

```

Output: - Enter two integers: 17 5
Remainder = 2

9. Write a C program to input a number and calculate its square.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int n, square;
clrscr();
printf("Enter a number: ");
scanf("%d", &n);
square = n * n;
printf("Square = %d", square);
getch();
}

Output: - Enter a number: 5
Square = 25

10. Write a C program to input a number and calculate its cube.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int n, cube;
clrscr();
printf("Enter a number: ");
scanf("%d", &n);
cube = n * n * n;
printf("Cube = %d", cube);
getch();
}

Output: - Enter a number: 5
Cube = 125

11. Write a C program to input length and breadth and calculate the area of a rectangle.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int length, breadth, area;
clrscr();
printf("Enter length: ");
scanf("%d", &length);
printf("Enter breadth: ");
scanf("%d", &breadth);
area = length * breadth;
printf("Area of Rectangle = %d", area);
getch();
}

```
}
```

Output: - Enter length: 10
Enter breadth: 5
Area of Rectangle = 50

12. Write a C program to input length and breadth and calculate the perimeter of a rectangle.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int length, breadth, perimeter;
clrscr();
printf("Enter length: ");
scanf("%d", &length);
printf("Enter breadth: ");
scanf("%d", &breadth);
perimeter = 2 * (length + breadth);
printf("Perimeter of Rectangle = %d", perimeter);
getch();
}

Output: - Enter length: 10
Enter breadth: 5
Perimeter of Rectangle = 30

13. Write a C program to input the radius and calculate the area of a circle.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
float radius, area;
float pi = 3.14;
clrscr();
printf("Enter radius: ");
scanf("%f", &radius);
area = pi * radius * radius;
printf("Area of Circle = %.2f", area);
getch();
}

Output: - Enter radius: 5
Area of Circle = 78.50

14. Write a C program to input principal, rate and time and calculate simple interest.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
float p, r, t, si;
clrscr();
printf("Enter Principal: ");
scanf("%f", &p);
printf("Enter Rate: ");
scanf("%f", &r);
printf("Enter Time: ");
scanf("%f", &t);
si = (p * r * t) / 100;
printf("Simple Interest = %f", si);
getch();
}

Output: - Enter Principal: 10000
Enter Rate: 5
Enter Time: 2
Simple Interest = 1000.00

15. Write a C program to input three numbers and calculate their average.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
float a, b, c, average;
clrscr();
printf("Enter three numbers: ");
scanf("%f%f%f", &a, &b, &c);
average = (a + b + c) / 3;
printf("Average = %.2f", average);
getch();
}

Output: - Enter three numbers: 10 20 30
Average = 20.00

16. Write a C program to input temperature in Celsius and convert it into Fahrenheit.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
float c, f;
clrscr();
printf("Enter temperature in Celsius: ");
scanf("%f", &c);
f = (c * 9 / 5) + 32;
printf("Temperature in Fahrenheit = %.2f", f);
getch();
}

Output: - Enter temperature in Celsius: 25
Temperature in Fahrenheit = 77.00

17. Write a C program to input two numbers and swap their values using a third variable.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int a, b, temp;
clrscr();
printf("Enter two numbers: ");
scanf("%d%d", &a, &b);
printf("Before Swapping: a = %d, b = %d\n", a, b);
temp = a;
a = b;
b = temp;
printf("After Swapping: a = %d, b = %d", a, b);
getch();
}

Output: - Enter two numbers: 10 20
Before Swapping: a = 10, b = 20
After Swapping: a = 20, b = 10

18. Write a C program to input marks of three subjects and calculate total marks and average.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
float m1, m2, m3, total, average;
clrscr();
printf("Enter marks of three subjects: ");
scanf("%f%f%f", &m1, &m2, &m3);
total = m1 + m2 + m3;
average = total / 3;
printf("Total Marks = %.2f\n", total);
printf("Average Marks = %.2f", average);
getch();
}

Output: - Enter marks of three subjects: 80 75 85
Total Marks = 240.00
Average Marks = 80.00

19. Write a C program to input basic salary, allowance and bonus and calculate the total salary.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
float basic, allowance, bonus, total;
clrscr();
printf("Enter Basic Salary: ");
scanf("%f", &basic);
printf("Enter Allowance: ");
scanf("%f", &allowance);
printf("Enter Bonus: ");
scanf("%f", &bonus);
total = basic + allowance + bonus;
printf("Total Salary = %.2f", total);
getch();
}

Output: - Enter Basic Salary: 20000
Enter Allowance: 3000
Enter Bonus: 2000
Total Salary = 25000.00

20. Write a C program to declare an integer, float, character and double variable and display their values.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int a = 10;
float b = 12.5;
char c = 'A';
double d = 123.4567;
clrscr();

printf("Integer = %d\n", a);
printf("Float = %.2f\n", b);
printf("Character = %c\n", c);
printf("Double = %lf", d);
getch();
}

```
}
```

Output: - Integer = 10
Float = 12.50
Character = A
Double = 123.456700

21. Write a C program to input a character from the user and display it.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
char ch;
clrscr();
printf("Enter a character: ");
scanf("%c", &ch);
printf("You entered: %c", ch);
getch();
}

Output: - Enter a character: A
You entered: A

22. Write a C program to input two numbers and display their addition, subtraction, multiplication, division and remainder.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int a, b;
clrscr();
printf("Enter two numbers: ");
scanf("%d%d", &a, &b);
printf("Addition = %d\n", a + b);
printf("Subtraction = %d\n", a - b);
printf("Multiplication = %d\n", a * b);
printf("Division = %d\n", a / b);
printf("Remainder = %d", a % b);
getch();
}

Output: - Enter two numbers: 20 5
Addition = 25
Subtraction = 15
Multiplication = 100
Division = 4
Remainder = 0

23. Write a C program to input a number, increment it by 1 and display the result.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int n;
clrscr();
printf("Enter a number: ");
scanf("%d", &n);
n++;
printf("After Increment = %d", n);
}

```
    getch();  
}
```

Output: - Enter a number: 10
After Increment = 11

24. Write a C program to input a number, decrement it by 1 and display the result.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
 int n;
 clrscr();
 printf("Enter a number: ");
 scanf("%d", &n);
 n--;
 printf("After Decrement = %d", n);
 getch();
}

Ans: - Enter a number: 10
After Decrement = 9

25. Write a C program to input values of a, b and c and calculate $a + b * c$.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
 int a, b, c, result;
 clrscr();
 printf("Enter a, b and c: ");
 scanf("%d%d%d", &a, &b, &c);
 result = a + b * c;
 printf("Result = %d", result);
 getch();
}

Output: - Enter a, b and c: 10 5 2
Result = 20

26. Write a C program to input values of a, b and c and calculate $(a + b) * c$.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
 int a, b, c, result;
 clrscr();
 printf("Enter a, b and c: ");
 scanf("%d%d%d", &a, &b, &c);
 result = (a + b) * c;
 printf("Result = %d", result);
 getch();
}

Output: - Enter a, b and c: 10 5 2
Result = 30

27. Write a C program to input the price and quantity of a product and calculate the total price.

Ans: - #include <stdio.h>


```

#include <conio.h>
void main()
{
    float price, quantity, total;
    clrscr();
    printf("Enter Price: ");
    scanf("%f", &price);
    printf("Enter Quantity: ");
    scanf("%f", &quantity);
    total = price * quantity;
    printf("Total Price = %.2f", total);
    getch();
}

```

Output: - Enter Price: 100
 Enter Quantity: 5
 Total Price = 500.00

28. Write a C program to input the current year and birth year and calculate the age.

Ans: - #include <stdio.h>
 #include <conio.h>
 void main()
 {
 int currentYear, birthYear, age;
 clrscr();
 printf("Enter Current Year: ");
 scanf("%d", ¤tYear);
 printf("Enter Birth Year: ");
 scanf("%d", &birthYear);
 age = currentYear - birthYear;
 printf("Age = %d years", age);
 getch();
 }

Output: - Enter Current Year: 2026
 Enter Birth Year: 2004
 Age = 22 years

29. Write a C program to input the number of minutes and convert them into hours and remaining minutes.

Ans: - #include <stdio.h>
 #include <conio.h>
 void main()
 {
 int minutes, hours, remaining;
 clrscr();
 printf("Enter total minutes: ");
 scanf("%d", &minutes);
 hours = minutes / 60;
 remaining = minutes % 60;
 printf("Hours = %d\n", hours);
 printf("Remaining Minutes = %d", remaining);
 getch();
 }

Output: - Enter total minutes: 150
 Hours = 2
 Remaining Minutes = 30

30. Write a C program to input marks of five subjects and calculate total marks and percentage.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
float m1, m2, m3, m4, m5;
float total, percentage;
clrscr();
printf("Enter marks of five subjects: ");
scanf("%f%f%f%f%f", &m1, &m2, &m3, &m4, &m5);
total = m1 + m2 + m3 + m4 + m5;
percentage = (total / 500) * 100;
printf("Total Marks = %.2f\n", total);
printf("Percentage = %.2f%%", percentage);
getch();
}

Output: - Enter marks of five subjects: 80 75 85 90 70
Total Marks = 400.00
Percentage = 80.00%

Q31. Write a C program to check whether a number is even or odd.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int n;
clrscr();
printf("Enter a number: ");
scanf("%d", &n);
if(n % 2 == 0)
printf("Even Number");
else
printf("Odd Number");
getch();
}

Output: - Enter a number: 15
Odd Number

Q32. Write a C program to check whether a number is positive, negative or zero.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int n;
clrscr();
printf("Enter a number: ");
scanf("%d", &n);
if(n > 0)
printf("Positive Number");
else if(n < 0)
printf("Negative Number");
else
printf("Zero");
getch();
}

Output: - Enter a number: -8

Negative Number

Q33. Write a C program to find the largest of two numbers.

```
Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
    int a, b;
    clrscr();
    printf("Enter two numbers: ");
    scanf("%d%d", &a, &b);
    if(a > b)
        printf("Largest = %d", a);
    else
        printf("Largest = %d", b);
    getch();
}
```

Output: - Enter Two number: - 10 48
Largest = 48

Q34. Write a C program to find the largest of three numbers.

```
Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
    int a, b, c;
    clrscr();
    printf("Enter three numbers: ");
    scanf("%d%d%d", &a, &b, &c);
    if(a >= b && a >= c)
        printf("Largest = %d", a);
    else if(b >= a && b >= c)
        printf("Largest = %d", b);
    else
        printf("Largest = %d", c);
    getch();
}
```

Output: - Enter three numbers: 25 70 40
Largest = 70

Q35. Write a C program to find the smallest of three numbers.

```
Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
    int a, b, c;
    clrscr();
    printf("Enter three numbers: ");
    scanf("%d%d%d", &a, &b, &c);
    if(a <= b && a <= c)
        printf("Smallest = %d", a);
    else if(b <= a && b <= c)
        printf("Smallest = %d", b);
    else
        printf("Smallest = %d", c);
    getch();
}
```

Output: - Enter three numbers: 25 70 10
Smallest = 10

**** Profit and Loss ****

Q36. Write a C program to calculate profit or loss.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
float cp, sp, amount;
clrscr();
printf("Enter Cost Price: ");
scanf("%f", &cp);
printf("Enter Selling Price: ");
scanf("%f", &sp);
if(sp > cp)
{
amount = sp - cp;
printf("Profit = %.2f", amount);
}
else if(cp > sp)
{
amount = cp - sp;
printf("Loss = %.2f", amount);
}
else
printf("No Profit No Loss");
getch();
}

Output: - Enter Cost Price: 500
Enter Selling Price: 650
Profit = 150.00

Q37. Write a C program to calculate profit percentage.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
float cp, sp, profit, percentage;
clrscr();
printf("Enter Cost Price: ");
scanf("%f", &cp);
printf("Enter Selling Price: ");
scanf("%f", &sp);
profit = sp - cp;
percentage = (profit / cp) * 100;
printf("Profit = %.2f\n", profit);
printf("Profit Percentage = %.2f%%", percentage);
getch();
}

Output: - Enter Cost Price: 500
Enter Selling Price: 600
Profit = 100.00
Profit Percentage = 20.00%

Even/Odd and Divisibility

Q38. Write a C program to print even numbers from 1 to 100.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int i;
clrscr();
for(i = 1; i <= 100; i++)
{
if(i % 2 == 0)
printf("%d ", i);
}
getch();
}

Output: - 2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50 52 54 56 58 60 62 64 66 68 70 72 74 76 78 80 82 84 86 88 90 92 94 96 98 100

Q39. Write a C program to print odd numbers from 1 to 100.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int i;
clrscr();
for(i = 1; i <= 100; i++)
{
if(i % 2 != 0)
printf("%d ", i);
}
getch();
}

Output: - 1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49 51 53 55 57 59 61 63 65 67 69 71 73 75 77 79 81 83 85 87 89 91 93 95 97 99

Q40. Write a C program to check whether a number is divisible by 5 and 11.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int n;
clrscr();
printf("Enter a number: ");
scanf("%d", &n);
if(n % 5 == 0 && n % 11 == 0)
printf("Number is divisible by 5 and 11");
else
printf("Number is not divisible by 5 and 11");
getch();
}

Output: - Enter a number: 55
Number is divisible by 5 and 11

Factorial and Series

Q41. Write a C program to find factorial of a number.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int n, i;
long int fact = 1;
clrscr();
printf("Enter a number: ");
scanf("%d", &n);
for(i = 1; i <= n; i++)
fact = fact * i;
printf("Factorial = %ld", fact);
getch();
}

Output: - Enter a number: 5
Factorial = 120

Q42. Write a C program to print multiplication table.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int n, i;
clrscr();
printf("Enter a number: ");
scanf("%d", &n);
for(i = 1; i <= 10; i++)
printf("%d x %d = %d\n", n, i, n*i);
getch();
}

Output: - Enter a number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

Q43. Write a C program to print numbers from 1 to N.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int n, i;
clrscr();
printf("Enter N: ");
scanf("%d", &n);
for(i = 1; i <= n; i++)
printf("%d ", i);
getch();
}

Output: - Enter N: 10
1 2 3 4 5 6 7 8 9 10

Q44. Write a C program to find sum of numbers from 1 to N.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int n, i, sum = 0;
clrscr();
printf("Enter N: ");
scanf("%d", &n);
for(i = 1; i <= n; i++)
sum = sum + i;
printf("Sum = %d", sum);
getch();
}

Output: - Enter N: 10
Sum = 55

Q45. Write a C program to print Fibonacci series.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int n, i, a = 0, b = 1, c;
clrscr();
printf("Enter number of terms: ");
scanf("%d", &n);
printf("Fibonacci Series:\n");
for(i = 1; i <= n; i++)
{
printf("%d ", a);
c = a + b;
a = b;
b = c;
}
getch();
}

Output: - Enter number of terms: 8
Fibonacci Series:
0 1 1 2 3 5 8 13

Reverse / Palindrome / Digits ★

Q46. Write a C program to reverse a number.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int n, rev = 0, rem;
clrscr();
printf("Enter a number: ");
scanf("%d", &n);
while(n != 0)
{
rem = n % 10;

```

        rev = rev * 10 + rem;
        n = n / 10;
    }
    printf("Reverse = %d", rev);
    getch();
}

```

Output: - Enter a number: 12345
Reverse = 54321

Q47. Write a C program to check whether a number is palindrome.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
 int n, original, rev = 0, rem;
 clrscr();
 printf("Enter a number: ");
 scanf("%d", &n);
 original = n;
 while(n != 0)
 {
 rem = n % 10;
 rev = rev * 10 + rem;
 n = n / 10;
 }
 if(original == rev)
 printf("Palindrome Number");
 else
 printf("Not a Palindrome Number");
 getch();
}

Output: - Enter a number: 121
Palindrome Number

Q48. Write a C program to count the digits of a number.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
 int n, count = 0;
 clrscr();
 printf("Enter a number: ");
 scanf("%d", &n);
 while(n != 0)
 {
 n = n / 10;
 count++;
 }
 printf("Number of digits = %d", count);
 getch();
}

Output: - Enter a number: 12345
Number of digits = 5

Q49. Write a C program to find sum of digits.

```
Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
    int n, sum = 0, rem;
    clrscr();
    printf("Enter a number: ");
    scanf("%d", &n);
    while(n != 0)
    {
        rem = n % 10;
        sum = sum + rem;
        n = n / 10;
    }
    printf("Sum of digits = %d", sum);
    getch();
}
```

Ans: - Enter a number: 12345
Sum of digits = 15

Q50. Write a C program to find product of digits.

```
Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
    int n, product = 1, rem;
    clrscr();
    printf("Enter a number: ");
    scanf("%d", &n);
    while(n != 0)
    {
        rem = n % 10;
        product = product * rem;
        n = n / 10;
    }
    printf("Product of digits = %d", product);
    getch();
}
```

Output: - Enter a number: 1234
Product of digits = 24

Armstrong Number ★

Q51. Write a C program to check whether a number is Armstrong or not.

```
Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
    int n, original, rem, sum = 0;
    clrscr();
    printf("Enter a number: ");
    scanf("%d", &n);
    original = n;
    while(n != 0)
    {
        rem = n % 10;
```

```

        sum = sum + rem * rem * rem;
        n = n / 10;
    }
    if(sum == original)
        printf("Armstrong Number");
    else
        printf("Not an Armstrong Number");
    getch();
}

```

Output: - Enter a number: 153
Armstrong Number

Logic: $153 = 1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$

Q52. Write a C program to print Armstrong numbers from 1 to 500.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int n, temp, rem, sum;
clrscr();
printf("Armstrong numbers from 1 to 500:\n");
for(n = 1; n <= 500; n++)
{
temp = n;
sum = 0;
while(temp != 0)
{
rem = temp % 10;
sum = sum + rem * rem * rem;
temp = temp / 10;
}
if(sum == n)
printf("%d ", n);
}
getch();
}

Output: - Armstrong numbers from 1 to 500:
1 153 370 371 407

Prime Numbers ★

Q53. Write a C program to check whether a number is prime.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int n, i, count = 0;
clrscr();
printf("Enter a number: ");
scanf("%d", &n);
for(i = 1; i <= n; i++)
{
if(n % i == 0)
count++;
}
if(count == 2)
printf("Prime Number");
else

```

        printf("Not a Prime Number");
    getch();
}

```

Output: - Enter a number: 17
Prime Number

Q54. Write a C program to print prime numbers from 1 to 100.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
 int n, i, count;
 clrscr();
 for(n = 2; n <= 100; n++)
 {
 count = 0;
 for(i = 1; i <= n; i++)
 {
 if(n % i == 0)
 count++;
 }
 if(count == 2)
 printf("%d ", n);
 }
 getch();
}

Output: - 2 3 5 7 11 13 17 19 23 29
31 37 41 43 47 53 59 61
67 71 73 79 83 89 97

HCF / LCM

Q55. Write a C program to find HCF/GCD of two numbers.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
 int a, b, i, hcf = 1;
 clrscr();
 printf("Enter two numbers: ");
 scanf("%d%d", &a, &b);
 for(i = 1; i <= a && i <= b; i++)
 {
 if(a % i == 0 && b % i == 0)
 hcf = i;
 }
 printf("HCF = %d", hcf);
 getch();
}

Output: - Enter two numbers: 12 18
HCF = 6

Q56. Write a C program to find LCM of two numbers.

Ans: - #include <stdio.h>
#include <conio.h>

```

void main()
{
    int a, b, max, lcm;
    clrscr();
    printf("Enter two numbers: ");
    scanf("%d%d", &a, &b);
    max = (a > b) ? a : b;
    while(1)
    {
        if(max % a == 0 && max % b == 0)
        {
            lcm = max;
            break;
        }
        max++;
    }
    printf("LCM = %d", lcm);
    getch();
}

```

Output: - Enter two numbers: 12 18
LCM = 36

Number Properties

Q57. Write a C program to check whether a number is perfect number.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
 int n, i, sum = 0;
 clrscr();
 printf("Enter a number: ");
 scanf("%d", &n);
 for(i = 1; i < n; i++)
 {
 if(n % i == 0)
 sum = sum + i;
 }
 if(sum == n)
 printf("Perfect Number");
 else
 printf("Not a Perfect Number");
 getch();
}

Output: - Enter a number: 28
Perfect Number

Q58. Write a C program to check whether a number is strong number.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
 int n, original, rem, i;
 long int sum = 0, fact;
 clrscr();
 printf("Enter a number: ");
 scanf("%d", &n);

```

original = n;
while(n != 0)
{
    rem = n % 10;
    fact = 1;
    for(i = 1; i <= rem; i++)
        fact = fact * i;
    sum = sum + fact;
    n = n / 10;
}
if(sum == original)
    printf("Strong Number");
else
    printf("Not a Strong Number");
getch();
}

```

Output: - Enter a number: 145
Strong Number

Logic: - $145 = 1! + 4! + 5! = 1 + 24 + 120 = 145$

Special Number Programs

Q59. Write a C program to check whether a number is automorphic.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int n, square, temp, flag = 1;
clrscr();
printf("Enter a number: ");
scanf("%d", &n);
square = n * n;
temp = n;
while(temp != 0)
{
if(temp % 10 != square % 10)
{
flag = 0;
break;
}
temp = temp / 10;
square = square / 10;
}
if(flag == 1)
printf("Automorphic Number");
else
printf("Not an Automorphic Number");
getch();
}

Output: - Enter a number: 25
Automorphic Number

Logic: - $25 * 25 = 625$ और 625 का last two digit 25 है।

Q60. Write a C program to check whether a number is neon number.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{

```

int n, square, sum = 0, rem;
clrscr();
printf("Enter a number: ");
scanf("%d", &n);
square = n * n;
while(square != 0)
{
    rem = square % 10;
    sum = sum + rem;
    square = square / 10;
}
if(sum == n)
    printf("Neon Number");
else
    printf("Not a Neon Number");
getch();
}

```

Output: - Enter a number: 9
Neon Number

Logic: - $9^2 = 81$
 $= 8 + 1 = 9$

Factor Related

Q61. Write a C program to find factors of a number.

Ans: - #include <stdio.h>
 #include <conio.h>
 void main()
 {
 int n, i;
 clrscr();
 printf("Enter a number: ");
 scanf("%d", &n);
 printf("Factors are: ");
 for(i = 1; i <= n; i++)
 {
 if(n % i == 0)
 printf("%d ", i);
 }
 getch();
 }

Output: - Enter a number: 12
Factors are: 1 2 3 4 6 12

Q62. Write a C program to count the factors of a number.

Ans: - #include <stdio.h>
 #include <conio.h>
 void main()
 {
 int n, i, count = 0;
 clrscr();
 printf("Enter a number: ");
 scanf("%d", &n);
 for(i = 1; i <= n; i++)
 {
 if(n % i == 0)
 count++;
 }
 printf("Number of factors = %d", count);
 }

```
    getch();  
}
```

Output: - Enter a number: 12
Number of factors = 6

Pattern Programs ★

Q63. Write a C program to print a square star pattern.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
 int i, j;
 clrscr();
 for(i = 1; i <= 5; i++)
 {
 for(j = 1; j <= 5; j++)
 {
 printf("* ");
 }
 printf("\n");
 }
 getch();
}

Output: - * * * * *
* * * * *
* * * * *
* * * * *
* * * * *

Q64. Write a C program to print right triangle star pattern.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
 int i, j;
 clrscr();
 for(i = 1; i <= 5; i++)
 {
 for(j = 1; j <= i; j++)
 printf("* ");
 printf("\n");
 }
 getch();
}

Output: - *
* *
* * *
* * * *
* * * * *

Q65. Write a C program to print inverted triangle pattern.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{

```

int i, j;
clrscr();
for(i = 5; i >= 1; i--)
{
    for(j = 1; j <= i; j++)
        printf("* ");
    printf("\n");
}
getch();
}

```

Output: - * * * * *

```

* * * *
* * *
* *
*

```

Q66. Write a C program to print number triangle.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
 int i, j;
 clrscr();
 for(i = 1; i <= 5; i++)
 {
 for(j = 1; j <= i; j++)
 printf("%d ", j);
 printf("\n");
 }
 getch();
}

Output: - 1

```

1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

Q67. Write a C program to print Floyd's triangle.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
 int i, j, n = 1;
 clrscr();
 for(i = 1; i <= 5; i++)
 {
 for(j = 1; j <= i; j++)
 {
 printf("%d ", n);
 n++;
 }
 printf("\n");
 }
 getch();
}

Output: - 1

2 3
4 5 6
7 8 9 10
11 12 13 14 15

Arrays ★

Q68. Write a C program to input and display elements of an array.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int a[5], i;
clrscr();
printf("Enter 5 elements:\n");
for(i = 0; i < 5; i++)
scanf("%d", &a[i]);
printf("Array elements:\n");
for(i = 0; i < 5; i++)
printf("%d ", a[i]);
getch();
}

Output: - Enter 5 elements:
10 20 30 40 50

Array elements:
10 20 30 40 50

Q69. Write a C program to find sum of array elements.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int a[5], i, sum = 0;
clrscr();
printf("Enter 5 elements:\n");
for(i = 0; i < 5; i++)
{
scanf("%d", &a[i]);
sum = sum + a[i];
}
printf("Sum = %d", sum);
getch();
}

Output: - Enter 5 elements:
10 20 30 40 50
Sum = 150

Q70. Write a C program to find largest element in an array.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
int a[5], i, largest;
clrscr();
printf("Enter 5 elements:\n");
for(i = 0; i < 5; i++)
scanf("%d", &a[i]);

```

largest = a[0];
for(i = 1; i < 5; i++)
{
    if(a[i] > largest)
        largest = a[i];
}
printf("Largest = %d", largest);
getch();
}

```

Output: - Enter 5 elements:
10 50 20 40 30
Largest = 50

Q71. Write a C program to find smallest element in an array.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
 int a[5], i, smallest;
 clrscr();
 printf("Enter 5 elements:\n");
 for(i = 0; i < 5; i++)
 scanf("%d", &a[i]);
 smallest = a[0];
 for(i = 1; i < 5; i++)
 {
 if(a[i] < smallest)
 smallest = a[i];
 }
 printf("Smallest = %d", smallest);
 getch();
}

Output: - Enter 5 elements:
10 50 5 40 30
Smallest = 5

Q72. Write a C program to search an element in an array.

Ans: - #include <stdio.h>
#include <conio.h>
void main()
{
 int a[5], i, search, found = 0;
 clrscr();
 printf("Enter 5 elements:\n");
 for(i = 0; i < 5; i++)
 scanf("%d", &a[i]);
 printf("Enter element to search: ");
 scanf("%d", &search);
 for(i = 0; i < 5; i++)
 {
 if(a[i] == search)
 {
 found = 1;
 break;
 }
 }
 if(found == 1)

```

        printf("Element Found at position %d", i + 1);
    else
        printf("Element Not Found");
    getch();
}

```

Output: - Enter 5 elements:

10 20 30 40 50

Enter element to search: 30

Element Found at position 3

Sorting

Q73. Write a C program to sort an array in ascending order.

Ans: - #include <stdio.h>

#include <conio.h>

void main()

```

{
    int a[5], i, j, temp;
    clrscr();
    printf("Enter 5 elements:\n");
    for(i = 0; i < 5; i++)
        scanf("%d", &a[i]);
    for(i = 0; i < 5; i++)
    {
        for(j = i + 1; j < 5; j++)
        {
            if(a[i] > a[j])
            {
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }
        }
    }
    printf("Ascending Order:\n");
    for(i = 0; i < 5; i++)
        printf("%d ", a[i]);
    getch();
}

```

Output: - Enter 5 elements:

50 20 40 10 30

Ascending Order:

10 20 30 40 50

Q74. Write a C program to sort an array in descending order.

Ans: - #include <stdio.h>

#include <conio.h>

void main()

```

{
    int a[5], i, j, temp;
    clrscr();
    printf("Enter 5 elements:\n");
    for(i = 0; i < 5; i++)
        scanf("%d", &a[i]);
    for(i = 0; i < 5; i++)
    {
        for(j = i + 1; j < 5; j++)
        {

```

```

        if(a[i] < a[j])
        {
            temp = a[i];
            a[i] = a[j];
            a[j] = temp;
        }
    }
}
printf("Descending Order:\n");
for(i = 0; i < 5; i++)
    printf("%d ", a[i]);
getch();
}

```

Output: - Enter 5 elements:

50 20 40 10 30

Descending Order:

50 40 30 20 10

Strings ★

Q75. Write a C program to input and display a string.

Ans: - #include <stdio.h>
 #include <conio.h>
 void main()
 {
 char name[50];
 clrscr();
 printf("Enter your name: ");
 gets(name);
 printf("Your Name is: ");
 puts(name);
 getch();
 }

Output: - Enter your name: XYZ Kumar

Your Name is: XYZ Kumar

Q76. Write a C program to find length of a string.

Ans: - #include <stdio.h>
 #include <conio.h>
 #include <string.h>
 void main()
 {
 char str[50];
 clrscr();
 printf("Enter a string: ");
 gets(str);
 printf("Length = %d", strlen(str));
 getch();
 }

Output: - Enter a string: Hello

Length = 5

Q77. Write a C program to copy one string into another.

Ans: - #include <stdio.h>
 #include <conio.h>
 #include <string.h>

```

void main()
{
    char str1[50], str2[50];
    clrscr();
    printf("Enter a string: ");
    gets(str1);
    strcpy(str2, str1);
    printf("Copied String = %s", str2);
    getch();
}

```

Output: - Enter a string: Computer
Copied String = Computer

Q78. Write a C program to compare two strings.

Ans: - #include <stdio.h>
#include <conio.h>
#include <string.h>
void main()
{
 char str1[50], str2[50];
 clrscr();
 printf("Enter first string: ");
 gets(str1);
 printf("Enter second string: ");
 gets(str2);
 if(strcmp(str1, str2) == 0)
 printf("Strings are Equal");
 else
 printf("Strings are Not Equal");
 getch();
}

Output: - Enter first string: Hello
Enter second string: Hello
Strings are Equal

Q79. Write a C program to concatenate two strings.

Ans: - #include <stdio.h>
#include <conio.h>
#include <string.h>
void main()
{
 char str1[100], str2[50];
 clrscr();
 printf("Enter first string: ");
 gets(str1);
 printf("Enter second string: ");
 gets(str2);
 strcat(str1, str2);
 printf("Combined String = %s", str1);
 getch();
}

Output: - Enter first string: Hello
Enter second string: World
Combined String = HelloWorld